Minutes Of Meeting-1

Module: Machine Learning (ML_PCOM7E)

Project: Airbnb Business Analysis – Track 1

Team: Group B

Attentence: Naeema Alnaqbi, Fatema Ahmed Rashed Hassen Alshehhi, Jonas Besier, Kalifa Burton

Date: 23 February 2026

Format: Virtual Discussion (WhatsApp)

1. Purpose of Meeting

To finalise:

- The business question
- Analytical structure of the report
- Model selection and justification
- Visualisation plan
- Role allocation and deadlines
- Alignment with postgraduate grading criteria

2. Business Question

The team agreed on the following question:

“How can Airbnb optimise pricing strategies in NYC by predicting listing prices using property features and identifying distinct neighbourhood market segments?”

This question was selected for its strategic relevance, clarity, and alignment with Track 1 (Regression & Clustering).

3. Structural Refinement

Following team feedback, particularly regarding the weighting of methodology and visualisation sections, the team agreed to:

- Structure the methodology into:
    - Data Understanding & Preprocessing
    - Regression Modelling
    - Clustering & Segmentation
    - Limitations
- Provide explicit justification for modelling choices.
- Include interpretation of results in business terms.
- Ensure reproducibility through fixed random_state.

The team confirmed that the report must be analytical, not descriptive, and directly aligned with the marking rubric.

4. Methodological Decisions

Regression Models:

- Linear Regression (baseline)
- Ridge Regression (regularisation)
- Random Forest (non-linear benchmark)

Clustering:

- K-Means with feature scaling
- Elbow Method used to determine optimal K
- K = 4 selected

Performance evaluation metrics:

- R^2
- MAE
- RMSE

All modelling decisions were justified and linked to strategic implications.

5. Visualisation Plan

The team agreed to include:

1. Raw vs Log Price Distribution
2. Correlation Heatmap
3. Regression Performance Comparison Table
4. Residual Plot
5. Elbow Method Plot
6. Cluster Summary Table

All visuals will be clearly referenced within the methodology section.

6. Roles and Responsibilities

- Naeema – Project Lead & Technical Lead (modelling, clustering, visualisations, final integration)
- Abdulla – Business Context
- Fatema – Predictive Analytics & Regression Relevance
- Jonas – Classical ML Justification
- Kalifa – Business Implications

Roles are fixed to ensure accountability.

7. Deadlines

- Individual sections due: Friday evening
- Integrated draft circulated: Tuesday
- Final review before submission

8. Academic Alignment

The team confirmed that the report must demonstrate:

- Knowledge and Understanding
- Application of Knowledge
- Criticality
- Structured Presentation

The team aims to achieve distinction level through justified analysis, strategic interpretation, and professional presentation.

Minutes of Meeting – 2

Module: Machine Learning (ML_PCOM7E)

Project: Airbnb Business Analysis – Track 1

Team: Group B

Attentence: Naeema Alnaqbi, Fatema Ahmed Rashed Hassen Alshehhi, Jonas Besier, Kalifa Burton

Date: 8 March 2026

Format: Virtual Discussion (WhatsApp)

Purpose of Meeting

To review the final version of the project report before submission.

Discussion Points

- The team reviewed the final integrated report.

- The modelling results (Regression and Clustering) were confirmed.
- Visualisations and interpretation were checked for clarity.
- Minor wording adjustments were made to improve conciseness and alignment with the word limit.

Outcome

The team agreed that the report is ready for submission.

Next Steps

- Naeema will submit the final report.
- Each team member will complete the Peer Review Form individually.